Effect of decaffeination of coffee or tea on gastro-oesophageal reflux.

BACKGROUND: Coffee and tea are believed to cause gastro-oesophageal reflux; however, the effects of these beverages and of their major component, caffeine, have not been quantified. The aim of this study was to evaluate gastro-oesophageal reflux induced by coffee and tea before and after a decaffeination process, and to compare it with water and water-containing caffeine. METHODS: Three-hour ambulatory pH-metry was performed on 16 healthy volunteers, who received 300 ml of (i) regular coffee, decaffeinated coffee or tap water ($n = 16$), (ii) normal tea, decaffeinated tea, tap water, or coffee adapted to normal tea in caffeine concentration ($n = 6$), and (iii) caffeine-free and caffeine-containing water ($n = 8$) together with a standardized breakfast. RESULTS: Regular coffee induced a significant ($P < 0.05$) gastro-oesophageal reflux compared with tap water and normal tea, which were not different from each other. Decaffeination of coffee significantly ($P < 0.05$) diminished gastro-oesophageal reflux, whereas decaffeination of tea or addition of caffeine to water had no effect. Coffee adapted to normal tea in caffeine concentration significantly ($P < 0.05$) increased gastro-oesophageal reflux. CONCLUSIONS: Coffee, in contrast to tea, increases gastro-oesophageal reflux, an effect that is less pronounced after decaffeination. Caffeine does not seem to be responsible for gastro-oesophageal reflux which must be attributed to other components of coffee.